

Towards Real-World Blind Face Restoration with Generative Facial Prior

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<https://github.com/TencentARC/GFPGAN>

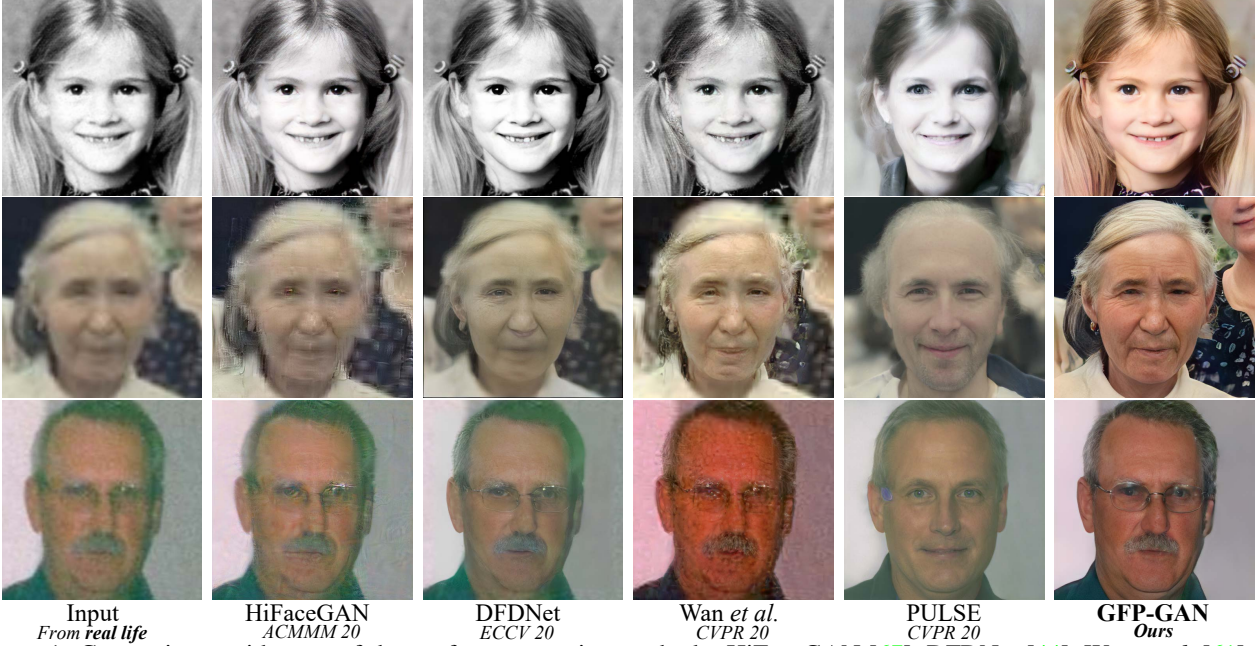


Figure 1: Comparisons with state-of-the-art face restoration methods: HiFaceGAN [67], DFDNet [44], Wan *et al.* [61] and PULSE [52] on the real-world low-quality images. While previous methods struggle to restore faithful facial details or retain face identity, our proposed GFP-GAN achieves a good balance of realness and fidelity with much fewer artifacts. In addition, the powerful generative facial prior allows us to perform restoration and color enhancement jointly. (Zoom in for best view)

Abstract

Blind face restoration usually relies on facial priors, such as facial geometry prior or reference prior, to restore realistic and faithful details. However, very low-quality inputs cannot offer accurate geometric prior while high-quality references are inaccessible, limiting the applicability in real-world scenarios. In this work, we propose GFP-GAN that leverages rich and diverse priors encapsulated in a pretrained face GAN for blind face restoration. This Generative Facial Prior (GFP) is incorporated into the face restoration process via spatial feature transform layers, which allow our method to achieve a good balance of realness and fidelity. Thanks to the powerful generative facial prior and delicate designs, our GFP-GAN could jointly restore facial details and enhance colors with just a single forward pass, while GAN inversion methods require image-specific optimization at inference. Extensive experiments show that our method achieves superior performance to prior art on both synthetic and real-world datasets.

1. Introduction

Blind face restoration aims at recovering high-quality faces from the low-quality counterparts suffering from unknown degradation, such as low-resolution [13, 48, 9], noise [71], blur [39, 58], compression artifacts [12], *etc.* When applied to real-world scenarios, it becomes more challenging, due to more complicated degradation, diverse poses and expressions. Previous works [9, 69, 6] typically exploit face-specific priors in face restoration, such as facial landmarks [9], parsing maps [6, 9], facial component heatmaps [69], and show that those *geometry facial priors* are pivotal to recover accurate face shape and details. However, those priors are usually estimated from input images and inevitably degrades with very low-quality inputs in the real world. In addition, despite their semantic guidance, the above priors contain limited texture information for restoring facial details (*e.g.*, eye pupil).

Another category of approaches investigates *reference priors*, *i.e.*, high-quality guided faces [46, 45, 11] or facial

面向现实世界的盲人脸复原研究——基于生成面部先验方法

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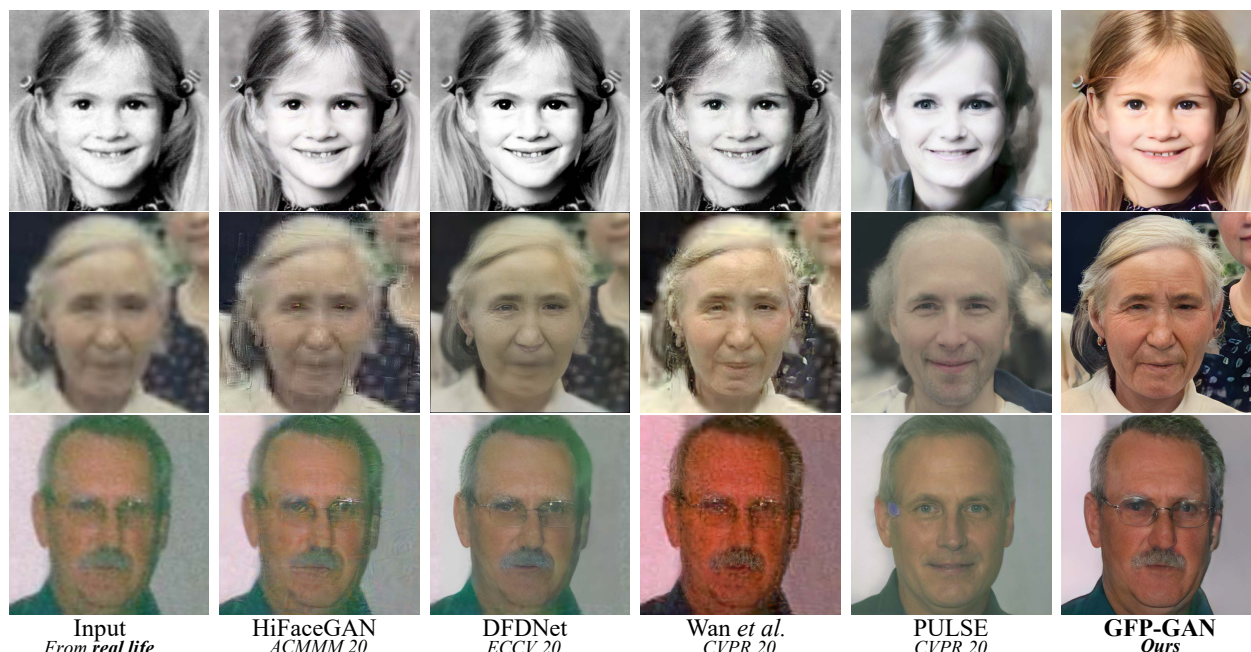


图1: 与最先进的面部恢复方法的比较: HiFaceGAN [67]、DFDNet [44]、Wan *et al.* [61] 和 PULSE [52] 在真实世界低质量图像上的表现。虽然之前的方法难以恢复真实的面部细节或保留人脸身份, 我们的提出的GFP-GAN在逼真度和保真度之间取得了良好的平衡, 并且生成的伪影显著减少。此外, 强大的生成式面部先验使我们能够同时进行恢复和色彩增强。(放大查看最佳效果)

摘要

Blind face restoration usually relies on facial priors, such as facial geometry prior or reference prior, to restore realistic and faithful details. However, very low-quality inputs cannot offer accurate geometric prior while high-quality references are inaccessible, limiting the applicability in real-world scenarios. In this work, we propose GFP-GAN that leverages rich and diverse priors encapsulated in a pretrained face GAN for blind face restoration. This Generative Facial Prior (GFP) is incorporated into the face restoration process via spatial feature transform layers, which allow our method to achieve a good balance of realness and fidelity. Thanks to the powerful generative facial prior and delicate designs, our GFP-GAN could jointly restore facial details and enhance colors with just a single forward pass, while GAN inversion methods require image-specific optimization at inference. Extensive experiments show that our method achieves superior performance to prior art on both synthetic and real-world datasets.

1. 引言

盲脸复原旨在从遭受未知退化(如低分辨率[13, 48, 9]、噪声[71]、模糊[39, 58]、压缩伪影[12])的低质量人脸图像中恢复高质量的人脸。当应用于实际场景时, 由于更复杂的退化情况以及多样的姿态和表情, 任务变得更加具有挑战性。以往的工作[9, 69, 6]通常利用特定于面部的先验知识进行面部复原, 例如面部标志点[9]、解析图[6, 9]、面部组件热图[69], 并表明这些方法对于恢复准确的人脸形状和细节至关重要。然而, 这些先验知识通常是根据输入图像估计得出的, 在实际中面对非常低质量的输入时不可避免地会退化。此外, 尽管它们提供了语义指导, 但上述先验知识包含有限的纹理信息用于恢复面部细节(如瞳孔)。

另一种方法类别研究的是高质量引导人脸[reference priors, i.e.]或面部[46, 45, 11]。

component dictionaries [44], to generate realistic results and alleviate the dependency on degraded inputs. However, the inaccessibility of high-resolution references limits its practical applicability, while the limited capacity of dictionaries restricts its diversity and richness of facial details.

In this study, we leverage *Generative Facial Prior* (GFP) for real-world blind face restoration, *i.e.*, the prior implicitly encapsulated in pretrained face Generative Adversarial Network (GAN) [18] models such as StyleGAN [35, 36]. These face GANs are capable of generating faithful faces with a high degree of variability, and thereby providing rich and diverse priors such as geometry, facial textures and colors, making it possible to jointly restore facial details and enhance colors (Fig. 1). However, it is challenging to incorporate such generative priors into the restoration process. Previous attempts typically use GAN inversion [19, 54, 52]. They first ‘invert’ the degraded image back to a latent code of the pretrained GAN, and then conduct expensive image-specific optimization to reconstruct images. Despite visually realistic outputs, they usually produce images with low fidelity, as the low-dimension latent codes are insufficient to guide accurate restoration.

To address these challenges, we propose the GFP-GAN with delicate designs to achieve a good balance of realness and fidelity in a single forward pass. Specifically, GFP-GAN consists of a degradation removal module and a pretrained face GAN as facial prior. They are connected by a direct latent code mapping, and several Channel-Split Spatial Feature Transform (CS-SFT) layers in a coarse-to-fine manner. The proposed CS-SFT layers perform *spatial* modulation on a split of features and leave the left features to directly pass through for better information preservation, allowing our method to effectively incorporate generative prior while retraining high fidelity. Besides, we introduce facial component loss with local discriminators to further enhance perceptual facial details, while employing identity preserving loss to further improve fidelity.

We summarize the contributions as follows. (1) We leverage rich and diverse generative facial priors for blind face restoration. Those priors contain sufficient facial textures and color information, allowing us to jointly perform face restoration and color enhancement. (2) We propose the GFP-GAN framework with delicate designs of architectures and losses to incorporate generative facial prior. Our GFP-GAN with CS-SFT layers achieves a good balance of fidelity and texture faithfulness in a single forward pass. (3) Extensive experiments show that our method achieves superior performance to prior art on both synthetic and real-world datasets.

2. Related Work

Image Restoration typically includes super-resolution [13, 48, 60, 49, 74, 68, 22, 50], denoising [71, 42, 26], deblurring [65, 39, 58] and compression removal [12, 21].

To achieve visually-pleasing results, generative adversarial network [18] is usually employed as loss supervisions to push the solutions closer to the natural manifold [41, 57, 64, 7, 14], while our work attempts to leverage the pretrained face GANs as generative facial priors (GFP).

Face Restoration. Based on general face hallucination [5, 30, 66, 70], two typical face-specific priors: geometry priors and reference priors, are incorporated to further improve the performance. The geometry priors include facial landmarks [9, 37, 77], face parsing maps [58, 6, 9] and facial component heatmaps [69]. However, 1) those priors require estimations from low-quality inputs and inevitably degrades in real-world scenarios. 2) They mainly focus on geometry constraints and may not contain adequate details for restoration. Instead, our employed GFP does not involve an explicit geometry estimation from degraded images, and contains adequate textures inside its pretrained network.

Reference priors [46, 45, 11] usually rely on reference images of the same identity. To overcome this issue, DFDNet [44] suggests to construct a face dictionary of each component (*e.g.*, eyes, mouth) with CNN features to guide the restoration. However, DFDNet mainly focuses on components in the dictionary and thus degrades in the regions beyond its dictionary scope (*e.g.*, hair, ears and face contour), instead, our GFP-GAN could treat faces as a whole to restore. Moreover, the limited size of dictionary restricts its diversity and richness, while the GFP could provide rich and diverse priors including geometry, textures and colors.

Generative Priors of pretrained GANs [34, 35, 36, 3] is previously exploited by GAN inversion [1, 76, 54, 19], whose primary aim is to find the closest latent codes given an input image. PULSE [52] iteratively optimizes the latent code of StyleGAN [35] until the distance between outputs and inputs is below a threshold. mGANprior [19] attempts to optimize multiple codes to improve the reconstruction quality. However, these methods usually produce images with low fidelity, as the low-dimension latent codes are insufficient to guide the restoration. In contrast, our proposed CS-SFT modulation layers enable prior incorporation on multi-resolution spatial features to achieve high fidelity. Besides, expensive iterative optimization is not required in our GFP-GAN during inference.

Channel Split Operation is usually explored to design compact models and improve model representation ability. MobileNet [28] proposes depthwise convolutions and GhostNet [23] splits the convolutional layer into two parts and uses fewer filters to generate intrinsic feature maps. Dual path architecture in DPN [8] enables feature re-usage and new feature exploration for each path, thus improving its representation ability. A similar idea is also employed in super-resolution [75]. Our CS-SFT layers share the similar spirits, but with different operations and purposes. We adopt spatial feature transform [63, 55] on one split and

组件字典 [44] 可以生成逼真的结果并缓解对质量下降的输入数据的依赖。然而，高分辨率参考数据的不可获取性限制了其实际应用，而字典容量的有限性则制约了面部细节的多样性和丰富度。

在这项研究中，我们利用 *Generative Facial Prior* (GFP) 进行实际场景中的盲人脸复原，即在预训练的面部生成对抗网络 (GAN) [18] 模型（如 StyleGAN [35, 36]）中隐含封装的先验知识。这些面部 GAN 能够生成具有高度变化性的逼真面孔，并提供丰富的多样化的先验信息，例如几何结构、面部纹理和颜色，从而使得同时复原面部细节并增强色彩成为可能（图1）。然而，将此类生成式先验融入到复原过程中颇具挑战性。以往的尝试通常采用 GAN 逆向映射 [19, 54, 52] 的方法。它们首先将退化图像“逆向”转换为预训练 GAN 中的潜在代码，然后进行昂贵的特定图像优化以重建图像。尽管输出视觉上非常逼真，但这些方法生成的图像通常保真度较低，因为低维潜在代码不足以引导准确复原。

为了应对这些挑战，我们提出了 GFP-GAN，并通过精心设计在单次前向传递中实现了真实性和保真度的良好平衡。具体来说，GFP-GAN 包含一个降质去除模块和一个预训练的面部 GAN 作为面部先验。它们通过直接隐码映射连接起来，并以从粗到细的方式嵌入了几层通道分割空间特征变换 (CS-SFT) 层。所提出的 CS-SFT 层对特征的一部分进行 *spatial* 调制，而让另一部分直接传递，从而更好地保留信息，使我们的方法能够在生成先验的同时保持高保真度。此外，我们引入了带有局部判别器的面部组件损失以进一步增强感知面部细节，并采用身份保持损失进一步提高保真度。

我们总结贡献如下：(1) 我们利用丰富多样的生成式面部先验进行盲人脸复原。这些先验包含了足够的面部纹理和色彩信息，使我们能够同时完成面部复原和色彩增强。(2) 我们提出了 GFP-GAN 框架，并通过精心设计的架构和损失函数来整合生成式面部先验。我们的 GFP-GAN 结合 CS-SFT 层，在单次前向传递中实现了保真度与纹理忠实性的良好平衡。(3) 大量实验表明，我们的方法在合成数据集和真实世界数据集中均优于现有技术。

2. 相关工作

图像恢复通常包括超分辨率处理 [13, 48, 60, 49, 74, 68, 22, 50]、去噪 [71, 42, 26]、去模糊 [65, 39, 58] 和解压缩处理 [12, 21]。

为了获得视觉上令人满意的结果，通常会使用生成对抗网络 [18] 作为损失监督，以将解决方案推向自然流形 [41, 57, 64, 7, 14]，而我们的工作则尝试利用预训练的面部 GAN 作为生成性面部先验 (GFP)。

面部修复。基于通用的面部超分辨率技术 [5, 30, 66, 70]，我们融合了两种典型的面部特定先验知识：几何先验和参考先验，以进一步提升性能。几何先验包括面部关键点 [9, 37, 77]、面部解析图 [58, 6, 9] 以及面部组件热力图 [69]。然而，1) 这些先验知识需要从低质量输入中进行估计，在实际场景中不可避免地会退化；2) 它们主要关注几何约束，可能不包含足够的细节用于修复。相反，我们采用的 GFP 不需要显式地从受损图像中估计几何信息，并且其预训练网络内部包含了充足的纹理细节。

参考先验 [46, 45, 11] 通常依赖于同一身份的参考图像。为了解决这一问题，DFD-Net [44] 建议构建每个面部组件（如眼睛、嘴巴）的 CNN 特征字典来指导恢复过程。然而，DFD-Net 主要关注字典中的组件，在超出其字典范围的区域（如头发、耳朵和脸部轮廓）性能会下降。相比之下，我们的 GFP-GAN 可以将整个面部视为一个整体来进行恢复。此外，受限于字典大小，其多样性和丰富性受到限制，而 GFP 能够提供包括几何形状、纹理和颜色在内的丰富多样的先验信息。

预训练 GAN 的生成先验 [34, 35, 36, 3] 此前已被 GAN 逆向工程 [1, 76, 54, 19] 利用，其主要目的是找到给定输入图像对应的最接近的潜在代码。PULSE [52] 通过迭代优化 StyleGAN [35] 的潜在代码，直到输出与输入之间的距离低于阈值为止。mGANprior [19] 尝试优化多个代码以提高重建质量。然而，这些方法通常会产生低保真度的图像，因为低维潜在代码不足以指导恢复过程。相比之下，我们提出的 CS-SFT 调制层能够在多分辨率空间特征上引入先验知识，从而实现高保真度。此外，在我们的 GFP-GAN 推理过程中不需要昂贵的迭代优化。

通道拆分操作通常被用于设计紧凑型模型并提升模型的表示能力。MobileNet [28] 提出了深度可分离卷积，而 GhostNet [23] 将卷积层分为两部分，并使用较少的滤波器生成固有特征图。DPN [8] 中的双路径架构允许每条路径重复利用特征并探索新特征，从而提升其表示能力。类似的思想也被应用于超分辨率领域 [75]。我们的 CS-SFT 层具有相似的精神，但采用了不同的操作和目的。我们在一个拆分部分采用空间特征变换 [63, 55]。

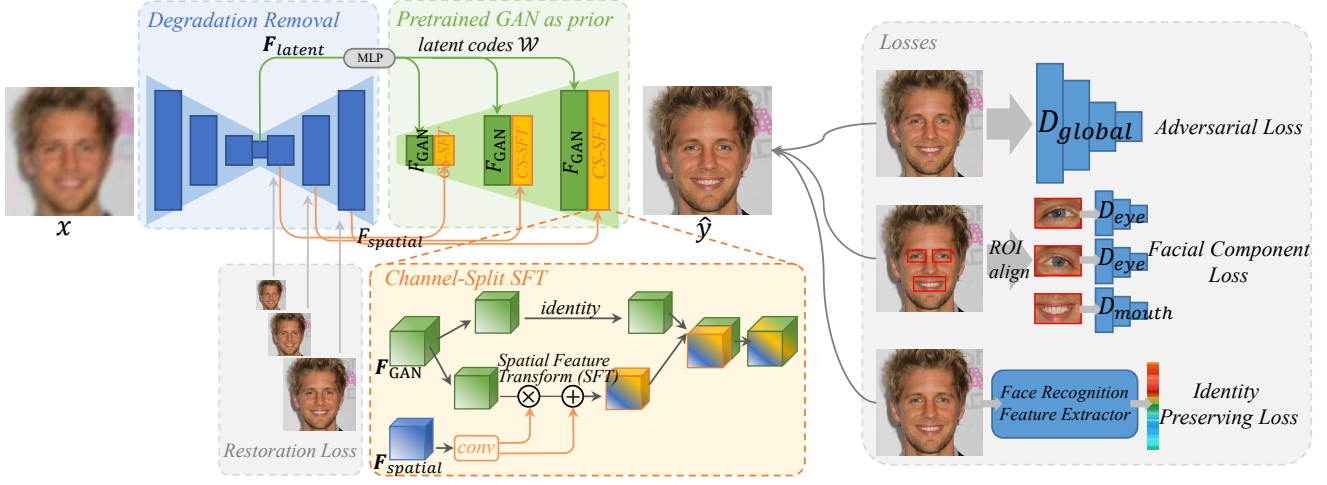


Figure 2: **Overview of GFP-GAN framework.** It consists of a degradation removal module (U-Net) and a pretrained face GAN as facial prior. They are bridged by a latent code mapping and several Channel-Split Spatial Feature Transform (CS-SFT) layers. During training, we employ 1) intermediate restoration losses to remove complex degradation, 2) Facial component loss with discriminators to enhance facial details, and 3) identity preserving loss to retain face identity.

leave the left split as identity to achieve a good balance of realness and fidelity.

Local Component Discriminators. Local discriminator is proposed to focus on local patch distributions [32, 47, 62]. When applied to faces, those discriminative losses are imposed on separate semantic facial regions [43, 20]. Our introduced facial component loss also adopts such designs but with a further style supervision based on the learned discriminative features.

3. Methodology

3.1. Overview of GFP-GAN

We describe GFP-GAN framework in this section. Given an input facial image x suffering from unknown degradation, the aim of blind face restoration is to estimate a high-quality image $\hat{y}$, which is as similar as possible to the ground-truth image y , in terms of realness and fidelity.

The overall framework of GFP-GAN is depicted in Fig. 2. GFP-GAN is comprised of a degradation removal module (U-Net) and a pretrained face GAN (such as StyleGAN2 [36]) as prior. They are bridged by a latent code mapping and several Channel-Split Spatial Feature Transform (CS-SFT) layers. Specifically, the degradation removal module is designed to remove complicated degradation, and extract two kinds of features, *i.e.* 1) latent features F_{latent} to map the input image to the closest latent code in StyleGAN2, and 2) multi-resolution spatial features $F_{spatial}$ for modulating the StyleGAN2 features.

After that, F_{latent} is mapped to intermediate latent codes $\mathcal{W}$ by several linear layers. Given the close latent code to the input image, StyleGAN2 could generate *intermediate convolutional features*, denoted by F_{GAN} . These features provide rich facial details captured in the weights of pre-

trained GAN. Multi-resolution features $F_{spatial}$ are used to spatially modulate the face GAN features F_{GAN} with the proposed CS-SFT layers in a coarse-to-fine manner, achieving realistic results while preserving high fidelity.

During training, except for the global discriminative loss, we introduce facial component loss with discriminators to enhance the perceptually significant face components, *i.e.*, eyes and mouth. In order to retrain identity, we also employ identity preserving guidance.

3.2. Degradation Removal Module

Real-world blind face restoration faces with complicated and severer degradation, which is typically a mixture of low-resolution, blur, noise and JPEG artifacts. The degradation removal module is designed to explicitly remove the above degradation and extract ‘clean’ features F_{latent} and $F_{spatial}$, alleviating the burden of subsequent modules. We adopt the U-Net [56] structure as our degradation remove module, as it could 1) increase receptive field for large blur elimination, and 2) generate multi-resolution features. The formulation is as follows:

$$F_{latent}, F_{spatial} = \text{U-Net}(x). \quad (1)$$

The latent features F_{latent} is used to map the input image to the closest latent code in StyleGAN2 (Sec. 3.3). The multi-resolution spatial features $F_{spatial}$ are used to modulate the StyleGAN2 features (Sec. 3.4).

In order to have an intermediate supervision for removing degradation, we employ the L1 restoration loss in each resolution scale in the early stage of training. Specifically, we also output images for each resolution scale of the U-Net decoder, and then restrict these outputs to be close to the pyramid of the ground-truth image.

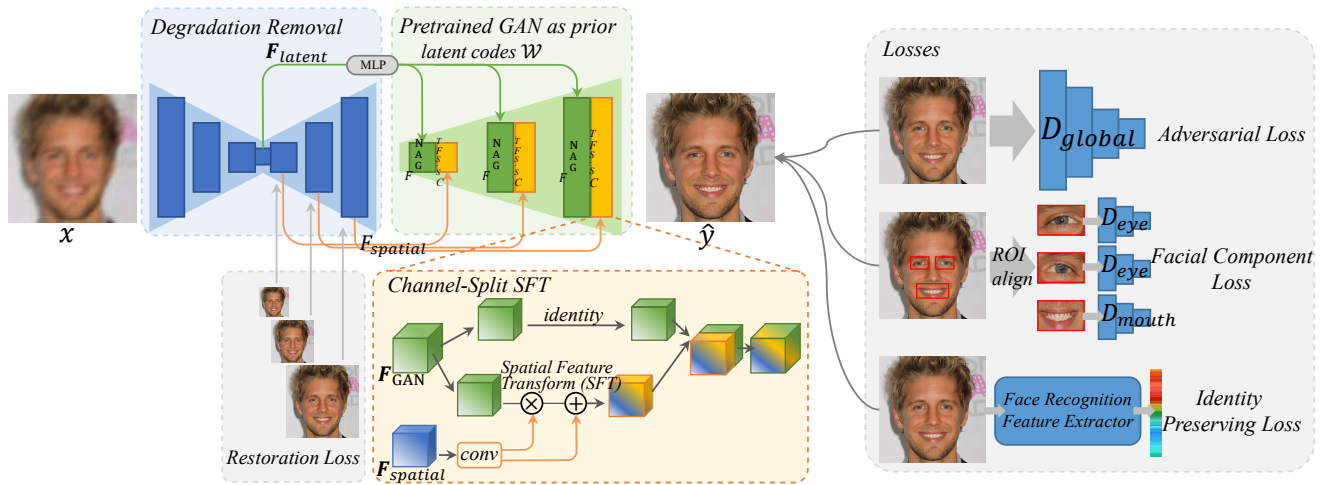


图2: GFP-GAN框架概述。该框架由一个降质去除模块（U-Net）和一个预训练的面部GAN作为面部先验组成。它们通过隐码映射和若干通道分割空间特征变换（CS-SFT）层连接起来。在训练过程中，我们采用1) 中间恢复损失来移除复杂降质，2) 带有判别器的面部组件损失以增强面部细节，3) 保持身份的损失以保留人脸身份。

保留左侧分裂为单位矩阵以实现真实性和保真度的良好平衡。

局部组件判别器。局部判别器被提出以关注局部补丁分布[32, 47, 62]。当应用于人脸时，这些判别性损失会被施加在不同的语义面部区域上[43, 20]。我们引入的面部组件损失也采用了类似的设计，但进一步加入了基于学习到的判别特征的风格监督。

3. 方法学

3.1 GFP-GAN 概述

在本节中，我们介绍了GFP-GAN框架。给定一张遭受未知退化的输入面部图像 x ，盲面部恢复的目标是估计出一张高质量的图像 $\hat{y}$ ，使其在真实感和保真度方面尽可能接近地面真相图像 y 。

GFP-GAN的整体框架如图2所示。GFP-GAN由一个降质去除模块（U-Net）和一个预训练的面部生成对抗网络（例如Style-GAN2 [36]）组成，后者作为先验条件。两者通过隐码映射和若干通道分割空间特征变换（CS-SFT）层连接起来。具体而言，降质去除模块旨在移除复杂的降质，并提取两类特征*i.e.*: 1) 隐式特征 F_{latent} ，用于将输入图像映射到StyleGAN2中最接近的隐码；2) 多分辨率空间特征 $F_{spatial}$ ，用于调节StyleGAN2的特征。

随后， F_{latent} 通过几个线性层映射到中间潜在码 W 。鉴于该潜在码与输入图像非常接近，StyleGAN2可以生成*intermediate convolutional features*，记为 F_{GAN} 。这些特征提供了面部细节在权重中的丰富捕捉。

训练后的生成对抗网络（GAN）。多分辨率特征 $F_{spatial}$ 被用于通过提出的CS-SFT层以由粗到精的方式对人脸GAN特征 F_{GAN} 进行空间调制，从而在保持高保真度的同时实现逼真的效果。

在训练过程中，除了全局判别损失外，我们还引入了面部组件损失，并使用判别器来增强感知上重要的面部组件，如眼睛和嘴巴。为了保持身份特征，我们也采用了身份保留引导。

3.2. 去劣化模块

现实世界中的盲人脸复原问题通常面临着复杂且严重的退化情况，这些退化通常是低分辨率、模糊、噪声和JPEG伪影的混合。降质去除模块旨在明确地移除上述退化，并提取“干净”的特征 F_{latent} 和 $F_{spatial}$ ，从而减轻后续模块的压力。我们采用U-Net [56]结构作为我们的降质去除模块，因为它可以1) 增大感受野以消除大规模模糊，2) 生成多分辨率特征。具体形式如下：

$$F_{latent}, F_{spatial} = \text{U-Net}(x). \quad (1)$$

潜特征 F_{latent} 用于将输入图像映射到StyleGAN2（第3.3节）中最接近的潜在代码。多分辨率空间特征 $F_{spatial}$ 用于调节StyleGAN2的特征（第3.4节）。

为了在训练早期阶段消除退化现象，我们在每个分辨率尺度上采用L1恢复损失进行中间监督。具体来说，我们还输出U-Net解码器中每个分辨率尺度的图像，并限制这些输出接近地面真相图像的金字塔结构。

3.3. Generative Facial Prior and Latent Code Mapping

A pre-trained face GAN captures a distribution over faces in its learned weights of convolutions, namely, generative prior [19, 54]. We leverage such pretrained face GANs to provide diverse and rich facial details for our task. A typical way of deploying generative priors is to map the input image to its closest latent codes Z , and then generate the corresponding output by a pretrained GAN [1, 76, 54, 19]. However, these methods usually require time-consuming iterative optimization for preserving fidelity. Instead of producing a final image directly, we generate *intermediate convolutional features* F_{GAN} of the closest face, as it contains more details and could be further modulated by input features for better fidelity (see Sec. 3.4).

Specifically, given the encoded vector F_{latent} of the input image (produced by the U-Net, Eq. 1), we first map it to intermediate latent codes $\mathcal{W}$ for better preserving semantic property *i.e.*, the intermediate space transformed from Z with several multi-layer perceptron layers (MLP) [76]. The latent codes $\mathcal{W}$ then pass through each convolution layer in the pre-trained GAN, and generate GAN features for *each resolution scale*.

$$\begin{aligned}\mathcal{W} &= \text{MLP}(F_{\text{latent}}), \\ F_{\text{GAN}} &= \text{StyleGAN}(\mathcal{W}).\end{aligned}\quad (2)$$

Discussion: Joint Restoration and Color Enhancement.

Generative models capture diverse and rich priors beyond realistic details and vivid textures. For instance, they also encapsulate *color* priors, which could be employed in our task for joint face restoration and color enhancement. Real-world face images, *e.g.*, old photos, usually have black-and-white color, vintage yellow color, or dim color. Lively color prior in generative facial prior allows us to perform color enhancement including colorization [72]. We believe the generative facial priors also incorporate conventional geometric priors [9, 69], 3D priors [16], *etc.* for restoration and manipulation.

3.4. Channel-Split Spatial Feature Transform

In order to better preserve fidelity, we further use the input spatial features F_{spatial} (produced by the U-Net, Eq. 1) to modulate the GAN features F_{GAN} from Eq. 2. Preserving spatial information from inputs is crucial for face restoration, as it usually requires local characteristics for fidelity preservation, and adaptive restoration at different spatial locations of a face. Therefore, we employ Spatial Feature Transform (SFT) [63], which generates affine transformation parameters for spatial-wise feature modulation, and has shown its effectiveness in incorporating other conditions in image restoration [63, 44] and image generation [55].

Specifically, at each resolution scale, we generate a pair of affine transformation parameters (α, β) from input fea-

tures F_{spatial} by several convolutional layers. After that, the modulation is carried out by scaling and shifting the GAN features F_{GAN} , formulated by:

$$\begin{aligned}\alpha, \beta &= \text{Conv}(F_{\text{spatial}}), \\ F_{\text{output}} &= \text{SFT}(F_{\text{GAN}}|\alpha, \beta) = \alpha \odot F_{\text{GAN}} + \beta.\end{aligned}\quad (3)$$

To achieve a better balance of realness and fidelity, we further propose channel-split spatial feature transform (CS-SFT) layers, which perform spatial modulation on part of the GAN features by input features F_{spatial} (contributing to fidelity) and leave the left GAN features (contributing to realness) to directly pass through, as shown in Fig. 2:

$$\begin{aligned}F_{\text{output}} &= \text{CS-SFT}(F_{\text{GAN}}|\alpha, \beta) \\ &= \text{Concat}[\text{Identity}(F_{\text{GAN}}^{\text{split}0}), \alpha \odot F_{\text{GAN}}^{\text{split}1} + \beta],\end{aligned}\quad (4)$$

where $F_{\text{GAN}}^{\text{split}0}$ and $F_{\text{GAN}}^{\text{split}1}$ are split features from F_{GAN} in channel dimension, and $\text{Concat}[\cdot, \cdot]$ denotes the concatenation operation.

As a result, CS-SFT enjoys the benefits of directly incorporating prior information and effective modulating by input images, thereby achieving a good balance between texture faithfulness and fidelity. Besides, CS-SFT could also reduce complexity as it requires fewer channels for modulation, similar to GhostNet [23].

We conduct channel-split SFT layers at each resolution scale, and finally generate a restored face $\hat{y}$.

3.5. Model Objectives

The learning objective of training our GFP-GAN consists of: 1) reconstruction loss that constraints the outputs $\hat{y}$ close to the ground-truth y , 2) adversarial loss for restoring realistic textures, 3) proposed facial component loss to further enhance facial details, and 4) identity preserving loss.

Reconstruction Loss. We adopt the widely-used L1 loss and perceptual loss [33, 41] as our reconstruction loss $\mathcal{L}_{\text{rec}}$, defined as follows:

$$\mathcal{L}_{\text{rec}} = \lambda_{\text{L1}} \|\hat{y} - y\|_1 + \lambda_{\text{per}} \|\phi(\hat{y}) - \phi(y)\|_1, \quad (5)$$

where ϕ is the pretrained VGG-19 network [59] and we use the $\{\text{conv1}, \dots, \text{conv5}\}$ feature maps before activation [64]. λ_{L1} and λ_{per} denote the loss weights of L1 and perceptual loss, respectively.

Adversarial Loss. We employ adversarial loss L_{adv} to encourage the GFP-GAN to favor the solutions in the natural image manifold and generate realistic textures. Similar to StyleGAN2 [36], logistic loss [18] is adopted:

$$L_{\text{adv}} = -\lambda_{\text{adv}} \mathbb{E}_{\hat{y}} \text{softplus}(D(\hat{y})) \quad (6)$$

where D denotes the discriminator and λ_{adv} represents the adversarial loss weight.

Facial Component Loss. In order to further enhance the perceptually significant face components, we introduce facial component loss with local discriminators for left eye,

3.3. 生成面部先验与潜在代码映射

预训练的面部GAN在其卷积权重中捕获了人脸分布，即生成先验[19,54]。我们利用这些预训练的面部GAN为我们的任务提供多样且丰富的面部细节。部署生成先验的一种典型方法是将输入图像映射到其最近似的潜在代码集 Z ，然后通过预训练的GAN生成相应的输出[1,76,54,19]。然而，这些方法通常需要耗时的迭代优化以保持保真度。相反，我们不是直接生成最终图像，而是生成最接近人脸的潜在代码集合

intermediate con-volutional features F_{GAN} ，因为它们包含更多细节，并且可以通过输入特征进一步调节以获得更好的保真度（参见第3.4节）。

具体而言，给定输入图像（由U-Net生成，见公式1）的编码向量 F_{latent} ，我们首先将其映射到中间潜在码 $\mathcal{W}$ 以更好地保留语义属性*i.e.*，该中间空间是从 Z 通过几个多层感知器层（MLP）[76]转换而来的。然后，这些潜在码 $\mathcal{W}$ 依次通过预训练的生成对抗网络（GAN）中的每个卷积层，并为*each resolution scale*生成GAN特征。

$$\begin{aligned}\mathcal{W} &= \text{MLP}(F_{latent}), \\ F_{GAN} &= \text{StyleGAN}(\mathcal{W}).\end{aligned}\quad (2)$$

讨论：联合修复与色彩增强。生成模型捕捉了超越现实细节和生动纹理的多样且丰富的先验知识。例如，它们还包含了*color*类先验知识，这些可以在我们的任务中用于联合人脸修复和色彩增强。实际世界中的面部图像，如*e.g.*老照片，通常为黑白、复古黄色或暗淡色调。生成模型中的生动色彩先验使我们能够进行包括着色在内的色彩增强[72]。我们认为生成的人脸先验还包含了传统的几何先验[9, 69]、3D先验[16]等*etc.*，用于修复和操作。

3.4. 通道拆分空间特征变换

为了更好地保持图像的保真度，我们进一步利用U-Net产生的输入空间特征 $F_{spatial}$ （见公式1）来调节生成对抗网络（GAN）的特征 F_{GAN} （见公式2）。在人脸修复中，保留来自输入的空间信息至关重要，因为这通常需要局部特性以保持保真度，并且要求在面部的不同空间位置进行自适应恢复。因此，我们采用空间特征变换（SFT）[63]，该方法生成用于空间维度上特征调节的仿射变换参数，在图像修复中已显示出其有效性，特别是在结合其他条件时[63, 44]以及在图像生成中的应用[55]。

具体而言，在每个分辨率尺度上，我们从输入特征生成一组仿射变换参数 (α, β) 。

通过几个卷积层进行特征提取 $F_{spatial}$ 。之后，通过对生成对抗网络（GAN）的特征 F_{GAN} 进行缩放和偏移来完成调制，具体公式为：

$$\begin{aligned}\alpha, \beta &= \text{Conv}(F_{spatial}), \\ F_{output} &= \text{SFT}(F_{GAN}|\alpha, \beta) = \alpha \odot F_{GAN} + \beta.\end{aligned}\quad (3)$$

为了更好地平衡真实感和保真度，我们进一步提出了通道分割空间特征变换（CS-SFT）层，该层通过输入特征对部分GAN特征进行空间调制以提升保真度，并让剩余的GAN特征（负责增强真实感的部分）直接传递，如图2所示：

$$\begin{aligned}F_{output} &= \text{CS-SFT}(F_{GAN}|\alpha, \beta) \\ &= \text{Concat}[\text{Identity}(F_{GAN}^{split0}), \alpha \odot F_{GAN}^{split1} + \beta],\end{aligned}\quad (4)$$

其中， F_{GAN}^{split0} 和 F_{GAN}^{split1} 是从 F_{GAN} 按通道维度分割出来的特征，而 $\text{Concat}[\cdot, \cdot]$ 表示拼接操作。

因此，CS-SFT 直接融合了先验信息，并通过输入图像有效调节，从而在纹理忠实度和保真度之间实现了良好的平衡。此外，由于 CS-SFT 调制所需的通道较少，类似于 GhostNet [23]，因此可以降低复杂性。

我们在每个分辨率尺度上进行通道分割的SFT层，最终生成一张恢复后的面部图像 $\hat{y}$ 。

3.5. 模型目标

我们的GFP-GAN训练目标包括：1) 重构损失，用于约束输出 $\hat{y}$ 接近真实值 y ；2) 对抗损失，用于恢复逼真的纹理；3) 提出的面部组件损失，进一步增强面部细节；4) 身份保持损失。

重构损失。我们采用广泛使用的L1损失和感知损失[33, 41]作为我们的重构损失，定义如下：

$$\mathcal{L}_{rec} = \lambda_{l1} \|\hat{y} - y\|_1 + \lambda_{per} \|\phi(\hat{y}) - \phi(y)\|_1, \quad (5)$$

其中， ϕ 是预训练的 VGG-19 网络 [59]，我们使用的是该网络在激活前的 $\{\text{conv1}, \dots, \text{conv5}\}$ 特征图 [64]。 λ_{l1} 和 λ_{per} 分别表示 L1 损失和感知损失的权重。

对抗损失。我们采用对抗损失 L_{adv} 来鼓励GFP-GAN倾向于自然图像流形中的解，并生成逼真的纹理。类似于StyleGAN2 [36]，采用了逻辑损失[18]：

$$\mathcal{L}_{adv} = -\lambda_{adv} \mathbb{E}_{\hat{y}} \text{softplus}(D(\hat{y})) \quad (6)$$

其中 D 表示判别器， λ_{adv} 表示对抗损失权重。

面部组件损失。为了进一步增强感知上重要的面部组件，我们引入了带有局部判别器的左眼面部组件损失。

right eye and mouth. As shown in Fig. 2, we first crop interested regions with ROI align [24]. For each region, we train separate and small local discriminators to distinguish whether the restore patches are real, pushing the patches close to the natural facial component distributions.

Inspired by [62], we further incorporate a feature style loss based on the learned discriminators. Different from previous feature matching loss with spatial-wise constraints [62], our feature style loss attempts to match the Gram matrix statistics [15] of real and restored patches. Gram matrix calculates the feature correlations and usually effectively captures texture information [17]. We extract features from multiple layers of the learned local discriminators and learn to match these Gram statistic of intermediate representations from the real and restored patches. Empirically, we found the feature style loss performs better than previous feature matching loss in terms of generating realistic facial details and reducing unpleasant artifacts.

The facial component loss is defined as follows. The first term is the discriminative loss [18] and the second term is the feature style loss:

$$\mathcal{L}_{comp} = \sum_{ROI} \lambda_{local} \mathbb{E}_{\hat{y}_{ROI}} [\log(1 - D_{ROI}(\hat{y}_{ROI}))] + \lambda_{fs} \|\text{Gram}(\psi(\hat{y}_{ROI})) - \text{Gram}(\psi(y_{ROI}))\|_1 \quad (7)$$

where ROI is region of interest from the component collection {left_eye, right_eye, mouth}. D_{ROI} is the local discriminator for each region. ψ denotes the multi-resolution features from the learned discriminators. λ_{local} and λ_{fs} represent the loss weights of local discriminative loss and feature style loss, respectively.

Identity Preserving Loss. We draw inspiration from [31] and apply identity preserving loss in our model. Similar to perceptual loss [33], we define the loss based on the feature embedding of an input face. Specifically, we adopt the pretrained face recognition ArcFace [10] model, which captures the most prominent features for identity discrimination. The identity preserving loss enforces the restored result to have a small distance with the ground truth in the compact deep feature space:

$$\mathcal{L}_{id} = \lambda_{id} \|\eta(\hat{y}) - \eta(y)\|_1, \quad (8)$$

where η represents face feature extractor, *i.e.* ArcFace [10] in our implementation. λ_{id} denotes the weight of identity preserving loss.

The overall model objective is a combination of the above losses:

$$\mathcal{L}_{total} = \mathcal{L}_{rec} + \mathcal{L}_{adv} + \mathcal{L}_{comp} + \mathcal{L}_{id}. \quad (9)$$

The loss hyper-parameters are set as follows: $\lambda_{l1} = 0.1$, $\lambda_{per} = 1$, $\lambda_{adv} = 0.1$, $\lambda_{local} = 1$, $\lambda_{fs} = 200$ and $\lambda_{id} = 10$.

4. Experiments

4.1. Datasets and Implementation

Training Datasets. We train our GFP-GAN on the FFHQ dataset [35], which consists of 70,000 high-quality images. We resize all the images to 512^2 during training.

Our GFP-GAN is trained on synthetic data that approximate to the real low-quality images and generalize to real-world images during inference. We follow the practice in [46, 44] and adopt the following degradation model to synthesize training data:

$$x = [(y \otimes k_\sigma) \downarrow_r + n_\delta]_{\text{JPEG}_q}. \quad (10)$$

The high quality image y is first convolved with Gaussian blur kernel k_σ followed by a downsampling operation with a scale factor r . After that, additive white Gaussian noise n_δ is added to the image and finally it is compressed by JPEG with quality factor q . Similar to [44], for each training pair, we randomly sample σ , r , δ and q from $\{0.2 : 10\}$, $\{1 : 8\}$, $\{0 : 15\}$, $\{60 : 100\}$, respectively. We also add color jittering during training for color enhancement.

Testing Datasets. We construct one synthetic dataset and three different **real** datasets with distinct sources. All these datasets have no overlap with our training dataset. We provide a brief introduction here.

- **CelebA-Test** is the synthetic dataset with 3,000 CelebA-HQ images from its testing partition [51]. The generation way is the same as that during training.

- **LFW-Test.** LFW [29] contains low-quality images **in the wild**. We group all the first image for each identity in the validation partition, forming 1711 testing images.

- **CelebChild-Test** contains 180 child faces of celebrities collected from **the Internet**. They are low-quality and many of them are black-and-white old photos.

- **WebPhoto-Test.** We crawled 188 low-quality photos **in real life from the Internet** and extracted 407 faces to construct the WebPhoto testing dataset. These photos have diverse and complicated degradation. Some of them are old photos with very severe degradation on both details and color.

Implementation. We adopt the pretrained StyleGAN2 [36] with 512^2 outputs as our generative facial prior. The channel multiplier of StyleGAN2 is set to one for compact model size. The UNet for degradation removal consists of seven downsamples and seven upsamples, each with a residual block [25]. For each CS-SFT layer, we use two convolutional layers to generate the affine parameters α and β respectively.

The training mini-batch size is set to 12. We augment the training data with horizontal flip and color jittering. We consider three components: left_eye, right_eye, mouth for face component loss as they are perceptually significant. Each component is cropped by ROI align [24] with face

右眼和嘴巴。如图2所示，我们首先使用ROI Align [24] 裁剪感兴趣区域。对于每个区域，我们训练单独且较小的局部判别器来区分恢复的补丁是否为真实数据，从而将这些补丁推向自然面部组件分布。

受文献[62]的启发，我们进一步引入了一种基于学习判别器的特征风格损失。与之前具有空间约束条件的特征匹配损失不同[62]，我们的特征风格损失尝试匹配真实和修复补丁的Gram矩阵统计量[15]。Gram矩阵计算特征相关性，并且通常能够有效捕捉纹理信息[17]。我们从学习到的局部判别器的多个层次中提取特征，并学会匹配这些来自真实和修复补丁中间表示的Gram统计量。经验表明，特征风格损失在生成逼真的面部细节和减少不愉快的伪影方面优于之前的特征匹配损失。

面部组件损失定义如下。第一项是判别损失[18]，第二项是特征风格损失。

$$\mathcal{L}_{comp} = \sum_{ROI} \lambda_{local} \mathbb{E}_{\hat{y}_{ROI}} [\log(1 - D_{ROI}(\hat{y}_{ROI}))] + \lambda_{fs} \|\text{Gram}(\psi(\hat{y}_{ROI})) - \text{Gram}(\psi(y_{ROI}))\|_1 \quad (7)$$

其中，ROI 是从组件集合

$\{\text{left_eye}, \text{right_eye}, \text{mouth}\}$ 中提取的兴趣区域。 D_{ROI} 表示每个兴趣区域的局部判别器。 ψ 表示来自学习到的判别器的多分辨率特征。 λ_{local} 和 λ_{fs} 分别表示局部判别损失和特征风格损失的权重。

身份保持损失。我们借鉴了[31]中的方法，在模型中应用了身份保持损失。类似于感知损失[33]，我们基于输入人脸的特征嵌入定义该损失。具体来说，我们采用了预训练的人脸识别ArcFace[10]模型，该模型捕捉了最具辨识度的身份特征。身份保持损失强制恢复结果在紧凑的深度特征空间中与真实值的距离较小：

$$\mathcal{L}_{id} = \lambda_{id} \|\eta(\hat{y}) - \eta(y)\|_1, \quad (8)$$

其中， η 表示面部特征提取器，*i.e.* 为 ArcFace [10]， λ_{id} 表示保真损失的权重。

总体模型目标是上述各项损失的组合：

$$\mathcal{L}_{total} = \mathcal{L}_{rec} + \mathcal{L}_{adv} + \mathcal{L}_{comp} + \mathcal{L}_{id}. \quad (9)$$

损失超参数设置如下： $\lambda_{l1} = 0.1$ ， $\lambda_{per} = 1$ ， $\lambda_{adv} = 0.1$ ， $\lambda_{local} = 1$ ， $\lambda_{fs} = 200$ ， $\lambda_{id} = 10$ 。

4. 实验

4.1 数据集与实现

训练数据集。我们在FFHQ数据集[35]上训练我们的GFP-GAN，该数据集包含70,000张高质量图像。我们将所有图像在训练时调整为512像素大小。

我们的GFP-GAN是在合成数据上进行训练的，这些数据近似于真实的低质量图像，并且在推理时能够泛化到真实世界中的图像。我们遵循[46, 44]中的做法，采用以下退化模型来生成训练数据：

$$x = [(y \otimes k_\sigma) \downarrow_r + n_\delta]_{\text{JPEG}_q}. \quad (10)$$

首先，高质量图像 y 与高斯模糊核 k_σ 进行卷积运算，随后进行缩放操作，缩放因子为 r 。之后，在图像中添加加性白高斯噪声 n_δ ，最后使用质量因子 q 对其进行JPEG 压缩。类似于 [44] 中的方法，对于每个训练对，我们分别从 $\{0.2 : 10\}$ 、 $\{1 : 8\}$ 、 $\{0 : 15\}$ 和 $\{60 : 100\}$ 中随机抽取 σ 、 r 、 δ 和 q 。在训练过程中，我们还添加颜色抖动以增强色彩效果。

测试数据集。我们构建了一个合成数据集和三个来源不同的真实数据集。所有这些数据集都与我们的训练数据集没有重叠。这里简要介绍如下。

- *CelebA-Test* 这是由其测试分区中的3,000张Celeb A-HQ图像组成的合成数据集[51]。生成方式与训练过程中相同。
- *LFW-Test*. LFW [29] 包含了质量较低的野外图像。我们将验证分区中每个身份的第一张图像进行分组，形成了1711张测试图像。
- *CelebChild-Test* 包含180张从互联网收集的名人的面部照片。这些照片质量较低，其中许多是黑白老照片。
- *WebPhoto-Test*. 我们从互联网上爬取了188张低质量的实物照片，并从中提取了407张人脸，构建了Web Photo测试数据集。这些照片具有多样且复杂的退化情况，其中一些是细节和色彩都严重退化的老照片。

实现。我们采用预训练的StyleGAN2[36]，其输出为512×49维度作为我们的生成面部先验模型。将StyleGAN2的通道乘数设置为1以减小模型尺寸。用于去除退化的UNet包含七次下采样和七次上采样操作，每次操作都配备一个残差块[25]。对于每个CS-SFT层，我们分别使用两个卷积层生成缩放和平移参数 α 和 β 。

训练的小批量大小设置为12。我们通过水平翻转和颜色抖动来增强训练数据。我们认为左眼、右眼和嘴巴这三个部分在面部组件损失中具有感知重要性，因此对每个部件使用ROI align [24]进行裁剪以提取人脸特征。

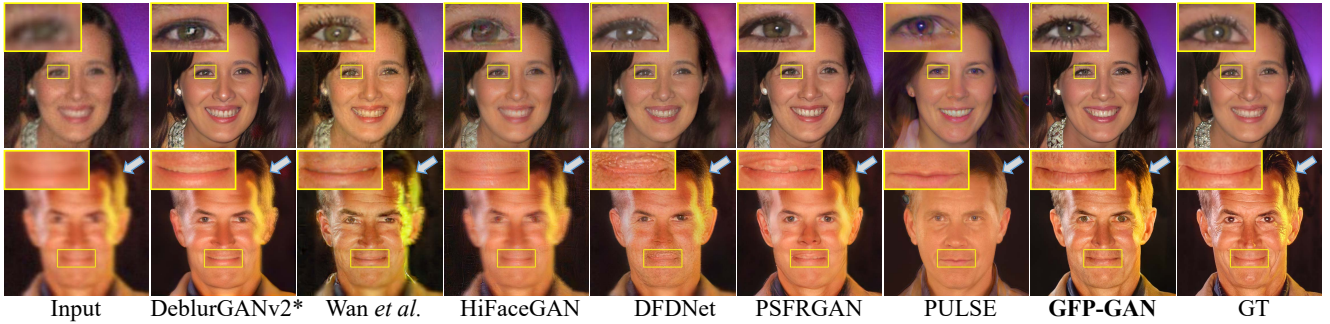


Figure 3: Qualitative comparison on the **CelebA-Test** for blind face restoration. Our GFP-GAN produces faithful details in eyes, mouth and hair. **Zoom in for best view.**

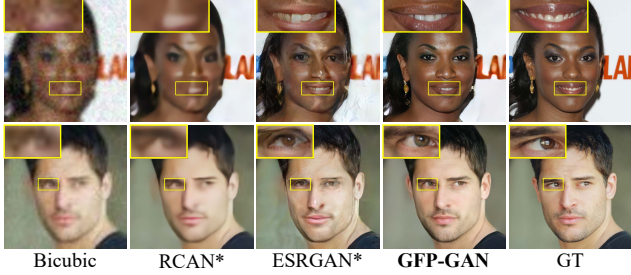


Figure 4: Comparison on the **CelebA-Test** for $\times 4$ face super-resolution. Our GFP-GAN restores realistic teeth and faithful eye gaze direction. **Zoom in for best view.**

landmarks provided in the origin training dataset. We train our model with Adam optimizer [38] for a total of 800k iterations. The learning rate was set to 2×10^{-3} and then decayed by a factor of 2 at the 700k-th, 750k-th iterations. We implement our models with the PyTorch framework and train them using four NVIDIA Tesla P40 GPUs.

4.2. Comparisons with State-of-the-art Methods

We compare our GFP-GAN with several state-of-the-art face restoration methods: HiFaceGAN [67], DFDNet [44], PSFRGAN [6], Super-FAN [4] and Wan *et al.* [61]. GAN inversion methods for face restoration: PULSE [52] and mGANprior [19] are also included for comparison. We also compare our GFP-GAN with image restoration methods: RCAN [74], ESRGAN [64] and DeblurGANv2 [40], and we finetune them on our face training set for fair comparisons. We adopt their *official* codes except for Super-FAN, for which we use a re-implementation.

For the evaluation, we employ the widely-used non-reference perceptual metrics: FID [27] and NIQE [53]. We also adopt pixel-wise metrics (PSNR and SSIM) and the perceptual metric (LPIPS [73]) for the CelebA-Test with Ground-Truth (GT). We measure the identity distance with angles in the ArcFace [10] feature embedding, where smaller values indicate closer identity to the GT.

Synthetic CelebA-Test. The comparisons are conducted under two settings: 1) blind face restoration whose inputs and outputs have the same resolution. 2) $4\times$ face super-resolution. Note that our method could take upsampled im-

Table 1: Quantitative comparison on **CelebA-Test** for blind face restoration. **Red** and **blue** indicates the best and the second best performance. ‘*’ denotes finetuning on our training set. Deg. represents the identity distance.

Methods	LPIPS↓	FID↓	NIQE↓	Deg.↓	PSNR↑	SSIM↑
Input	0.4866	143.98	13.440	47.94	25.35	0.6848
DeblurGANv2* [40]	0.4001	52.69	4.917	39.64	25.91	0.6952
Wan <i>et al.</i> [61]	0.4826	67.58	5.356	43.00	24.71	0.6320
HiFaceGAN [67]	0.4770	66.09	4.916	42.18	24.92	0.6195
DFDNet [44]	0.4341	59.08	4.341	40.31	23.68	0.6622
PSFRGAN [6]	0.4240	47.59	5.123	39.69	24.71	0.6557
mGANprior [19]	0.4584	82.27	6.422	55.45	24.30	0.6758
PULSE [52]	0.4851	67.56	5.305	69.55	21.61	0.6200
GFP-GAN (ours)	0.3646	42.62	4.077	34.60	25.08	0.6777
GT	0	43.43	4.292	0	∞	1

ages as inputs for face super-resolution.

The quantitative results for each setting are shown in Table 1 and Table 2. On both settings, GFP-GAN achieves the lowest LPIPS, indicating that our results is perceptually close to the ground-truth. GFP-GAN also obtain the lowest FID and NIQE, showing that the outputs have a close distance to the real face distribution and natural image distribution, respectively. Besides the perceptual performance, our method also retains better identity, indicated by the smallest degree in the face feature embedding. Note that 1) the lower FID and NIQE of our method than GT does not indicate that our performance is better than GT, as those ‘perceptual’ metrics are well correlated with the human-opinion-scores on a coarse scale, but *not always well correlated on a finer scale* [2]; 2) the pixel-wise metrics PSNR and SSIM are not correlation well with the subjective evaluation of human observers [2, 41] and our model is not good at these two metrics.

Qualitative results are presented in Fig. 3 and Fig. 4. 1) Thanks to the powerful generative facial prior, our GFP-GAN recovers faithful details in the eyes (pupils and eyelashes), teeth, *etc.* 2) Our method treats faces as whole in restoration and could also generate realistic hair, while previous methods that rely on component dictionaries (DFDNet) or parsing maps (PSFRGAN) fail to produce faithful hair textures (2nd row, Fig. 3). 3) GFP-GAN is capable of retaining fidelity, *e.g.*, it produces natural closed mouth

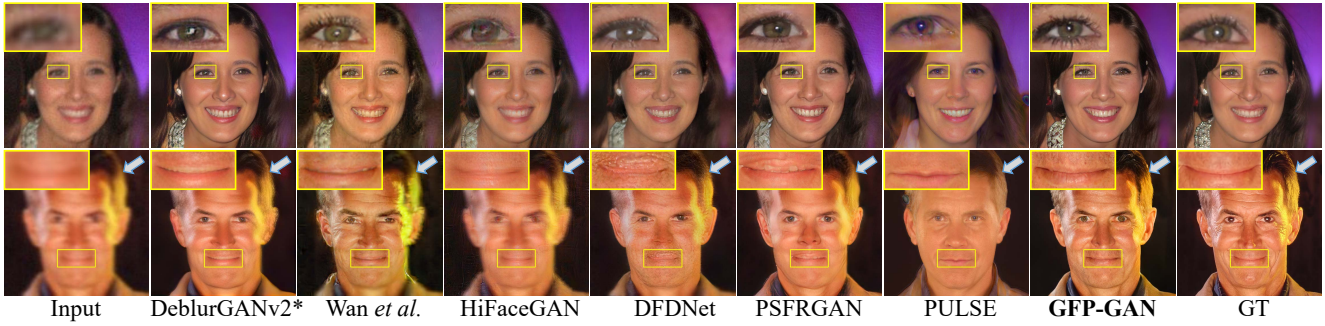


图3: 在CelebA-Test数据集上进行盲人脸恢复的定性比较。我们的GFP-GAN生成了眼睛、嘴巴和头发等部位忠实的细节。请放大查看以获得最佳效果。

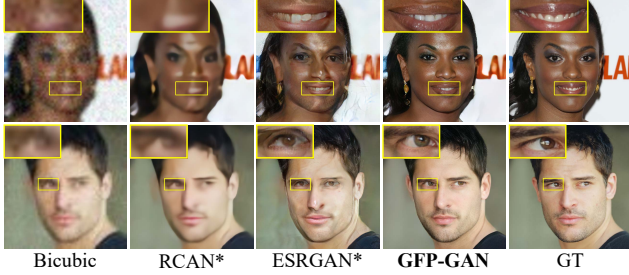


图4: 在CelebA-Test数据集上对比 $\times 4$ 人脸超分辨率的效果。我们的GFP-GAN能够恢复逼真的牙齿和准确的眼部视线方向。请放大查看以获得最佳效果。

我们在原始训练数据集中提供的地上训练模型。使用Adam优化器[38]进行总共80万次迭代的训练。学习率初始设置为 2×10^{-4} , 然后在第70万次和第75万次迭代时分别以2为因子衰减。我们使用PyTorch框架并在四块NVIDIA Tesla P40 GPU上进行模型训练。

4.2. 与最新方法的比较

我们将我们的GFP-GAN与几种最先进的面部恢复方法进行了比较: HiFaceGAN [67]、DFDNet [44]、PSFRGAN [6]、Super-FAN [4]和Wan *et al* [61]。用于面部恢复的GAN逆向方法还包括PULSE [52] 和 mGANprior [19], 这些也被纳入了比较范围。此外, 我们还将GFP-GAN与图像恢复方法进行了对比: RCAN [74]、ESRGAN [64]和DeblurGANv2 [40], 并在我们的面部训练集上对它们进行微调以确保公平的比较。除了Super-FAN外, 其余均采用其*official*代码, 对于Super-FAN则使用了重新实现的版本。

为了评估, 我们采用了广泛使用的无参考感知度量标准: FID[27]和NIQE[53]。对于具有真实标签 (GT) 的CelebA-Test数据集, 我们也使用了像素级度量指标 (PSNR和SSIM) 以及感知度量指标 (LPIPS[73])。我们通过ArcFace[10]特征嵌入中的角度来衡量身份距离, 其中较小的值表示与真实标签更接近的身份。

合成CelebA测试。比较在两种设置下进行: 1) 无参考人脸恢复, 输入和输出具有相同的分辨率。2) $4\times$ 人脸超分辨率。需要注意的是, 我们的方法可以处理上采样的图像。

表1: CelebA-Test数据集上盲人脸恢复的定量比较。红色和蓝色分别表示最佳和次佳性能。“*”表示在我们的训练集上进行了微调。Deg. 表示身份距离。

Methods	LPIPS↓	FID↓	NIQE↓	Deg.↓	PSNR↑	SSIM↑
Input	0.4866	143.98	13.440	47.94	25.35	0.6848
DeblurGANv2* [40]	<u>0.4001</u>	52.69	4.917	<u>39.64</u>	25.91	0.6952
Wan <i>et al.</i> [61]	0.4826	67.58	5.356	43.00	24.71	0.6320
HiFaceGAN [67]	0.4770	66.09	4.916	42.18	24.92	0.6195
DFDNet [44]	0.4341	59.08	<u>4.341</u>	40.31	23.68	0.6622
PSFRGAN [6]	0.4240	<u>47.59</u>	5.123	39.69	24.71	0.6557
mGANprior [19]	0.4584	82.27	6.422	55.45	24.30	0.6758
PULSE [52]	0.4851	67.56	5.305	69.55	21.61	0.6200
GFP-GAN (ours)	0.3646	42.62	4.077	34.60	25.08	0.6777
GT	0	43.43	4.292	0	∞	1

作为输入用于面部超分辨率。

表1和表2展示了每种设置下的定量结果。在两种设置下, GFP-GAN均获得了最低的LPIPS值, 表明我们的结果在感知上接近真实情况。此外, GFP-GAN还取得了最低的FID和NIQE值, 这说明其输出与真实的面部分布和自然图像分布分别具有较小的距离。除了感知性能外, 我们的方法还能更好地保留身份特征, 这一点从最小的脸部特征嵌入角度可以得到证明。需要注意的是: 1) 我们方法的FID和NIQE值低于GT, 并不意味着我们的表现优于GT, 因为这些“感知”指标与人类评分在粗略尺度上具有良好的相关性; 2) 像素级别的PSNR和SSIM度量并不很好地反映人类观察者的主观评价[2, 41], 且我们的模型在这两个指标上的表现不佳。

图3和图4展示了定性结果。1) 得益于强大的生成面部先验, 我们的GFP-GAN能够恢复眼睛 (瞳孔和睫毛)、牙齿等部位的忠实细节。2) 我们的方法在修复过程中将人脸视为整体, 并能生成逼真的头发, 而依赖于组件字典 (DFD-Net) 或解析图 (PSFRGAN) 的先前方法无法产生忠实的头发纹理 (图3第2行)。3) GFP-GAN能够保持保真度, 它还能生成自然闭合的嘴唇。



Figure 5: Qualitative comparisons on three **real-world** datasets. **Zoom in for best view.**

Table 2: Quantitative comparison on **CelebA-Test** for 4× face super-resolution. **Red** and **blue** indicates the best and the second best performance. “*” denotes finetuning on our training set. Deg. represents the identity distance.

Methods	LPIPS↓	FID↓	NIQE↓	Deg.↓	PSNR↑	SSIM↑
Bicubic	0.4834	148.87	10.767	49.60	25.377	0.6985
RCAN* [74]	0.4159	93.66	9.907	<u>38.45</u>	27.24	0.7533
ESRGAN* [64]	<u>0.4127</u>	<u>49.20</u>	<u>4.099</u>	<u>51.21</u>	23.74	0.6319
Super-FAN [4]	0.4791	139.49	10.828	49.14	25.28	<u>0.7033</u>
GFP-GAN (ours)	0.3653	42.36	4.078	34.67	25.04	0.6744
GT	0	43.43	4.292	0	∞	1

without forced addition of teeth as PSFRGAN does (2nd row, Fig. 3). And in Fig. 4, GFP-GAN also restores reasonable eye gaze direction.

Real-World LFW, CelebChild and WebPhoto-Test. To test the generalization ability, we evaluate our model on three different real-world datasets. The quantitative results are shown in Table 3. Our GFP-GAN achieves superior performance on all the three real-world datasets, showing its remarkable generalization capability. Although PULSE [52] could also obtain high perceptual quality (lower FID scores), it could not retain the face identity as

Table 3: Quantitative comparison on the *real-world* **LFW**, **CelebChild**, **WebPhoto**. **Red** and **blue** indicates the best and the second best performance. “*” denotes finetuning on our training set. Deg. represents the identity distance.

Dataset	LFW-Test		CelebChild		WebPhoto	
Methods	FID↓	NIQE↓	FID↓	NIQE↓	FID↓	NIQE↓
Input	137.56	11.214	144.42	9.170	170.11	12.755
DeblurGANv2* [40]	57.28	4.309	110.51	4.453	100.58	<u>4.666</u>
Wan et al. [61]	73.19	5.034	115.70	4.849	100.40	5.705
HiFaceGAN [67]	64.50	4.510	113.00	4.855	116.12	4.885
DFDNet [44]	62.57	<u>4.026</u>	111.55	<u>4.414</u>	100.68	5.293
PSFRGAN [6]	<u>51.89</u>	5.096	<u>107.40</u>	4.804	88.45	5.582
mGANprior [19]	73.00	6.051	126.54	6.841	120.75	7.226
PULSE [52]	64.86	5.097	102.74	5.225	86.45	5.146
GFP-GAN (ours)	49.96	3.882	111.78	4.349	<u>87.35</u>	4.144

shown in Fig 5.

The qualitative comparisons are shown in Fig. 5. GFP-GAN could jointly conduct face restoration and color enhancement for real-life photos with the powerful generative prior. Our method could produce plausible and realistic faces on complicated real-world degradation while other methods fail to recover faithful facial details or produces ar-



Figure 5: Qualitative comparisons on three **real-world** datasets. **Zoom in for best view.**

表2: CelebA-Test数据集上针对4×人脸超分辨率的定量比较。红色和蓝色分别表示最佳和次佳性能。“*”表示在我们的训练集上进行了微调。Deg. 表示身份距离。

Methods	LPIPS↓	FID↓	NIQE ↓	Deg.↓	PSNR↑	SSIM↑
Bicubic	0.4834	148.87	10.767	49.60	25.377	0.6985
RCAN* [74]	0.4159	93.66	9.907	<u>38.45</u>	27.24	0.7533
ESRGAN* [64]	<u>0.4127</u>	<u>49.20</u>	4.099	<u>51.21</u>	23.74	0.6319
Super-FAN [4]	0.4791	139.49	10.828	49.14	25.28	0.7033
GFP-GAN (ours)	0.3653	42.36	4.078	34.67	25.04	0.6744
GT	0	43.43	4.292	0	∞	1

无需像PSFRGAN那样强制添加牙齿（图3第2行）。在图4中，GFP-GAN也能合理恢复眼睛的视线方向。

现实世界中的LFW、CelebChild和WedPhoto-Test数据集。为了测试模型的泛化能力，我们在三个不同的真实世界数据集上对其进行了评估。定量结果见表3。我们的GFP-GAN在所有三个真实世界数据集中均表现出色，展示了其卓越的泛化能力。尽管PULSE [52]也能获得较高的感知质量（较低的FID分数），但它无法保持面部身份信息。

表3: *real-world* LFW、CelebChild和WebPhoto的定量比较。红色和蓝色分别表示最佳和次佳性能。“*”表示在我们的训练集上进行了微调。Deg. 表示身份距离。

Dataset	LFW-Test		CelebChild		WebPhoto	
Methods	FID↓	NIQE ↓	FID↓	NIQE ↓	FID↓	NIQE ↓
Input	137.56	11.214	144.42	9.170	170.11	12.755
DeblurGANv2* [40]	57.28	4.309	110.51	4.453	100.58	4.666
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PSFRGAN [6]	<u>51.89</u>	5.096	<u>107.40</u>	4.804	88.45	5.582
mGANprior [19]	73.00	6.051	126.54	6.841	120.75	7.226
PULSE [52]	64.86	5.097	102.74	5.225	86.45	5.146
GFP-GAN (ours)	49.96	3.882	111.78	4.349	<u>87.35</u>	4.144

如图5所示。

如图5所示，GFP-GAN能够联合进行真实生活照片中的面部修复和色彩增强，利用强大的生成先验。我们的方法能够在复杂的现实世界退化情况下产生合理且逼真面部图像，而其他方法则无法恢复忠实的面部细节或会产生不自然的结果。

Table 4: Ablation study results on **CelebA-Test** under blind face restoration.

Configuration	LPIPS↓	FID↓	NIQE↓	Deg.↓
Our GFP-GAN with SC-SFT	0.3646	42.62	4.077	34.60
a) No spatial modulation	0.550 (↑)	60.44 (↑)	4.183 (↑)	74.76 (↑)
b) Use SFT	0.387 (↑)	47.65 (↑)	4.146 (↑)	34.38 (↓)
c) w/o GFP	0.379 (↑)	48.47 (↑)	4.153 (↑)	35.04 (↑)
d) – Pyramid Restoration Loss	0.369 (↑)	45.17 (↑)	4.284 (↑)	35.50 (↑)

tifacts (especially in WebPhoto-Test in Fig 5). Besides the common facial components like eyes and teeth, GFP-GAN also perform better in hair and ears, as the GFP prior takes the whole face into consideration rather than separate parts. With SC-SFT layers, our model is capable of achieving high fidelity. As shown in the last row of Fig. 5, most previous methods fail to recover the closed eyes, while ours could successfully restore them with fewer artifacts.

4.3. Ablation Studies

CS-SFT layers. As shown in Table. 4 [configuration a)] and Fig. 6, when we remove spatial modulation layers, *i.e.*, only keep the latent code mapping without spatial information, the restored faces could not retain face identity even with identity-preserving loss (high LIPS score and large Deg.). Thus, the multi-resolution spatial features used in CS-SFT layers is vital to preserve fidelity. When we switch CS-SFT layers to simple SFT layers [configuration b) in Table. 4], we observe that 1) the perceptual quality degrades on all metrics and 2) it preserves stronger identity (smaller Deg.), as the input image features impose influence on all the modulated features and the outputs bias to the degraded inputs, thus leading to lower perceptual quality. By contrast, CS-SFT layers provide a good balance of realness and fidelity by modulating a split of features.

Pretrained GAN as GFP. Pretrained GAN provides rich and diverse features for restoration. A performance drop is observed if we do not use the generative facial prior, as shown in Table. 4 [configuration c)] and Fig. 6.

Pyramid Restoration Loss. Pyramid restoration loss is employed in the degradation removal module and strengthens the restoration ability for complicated degradation in the real world. Without this intermediate supervision, the multi-resolution spatial features for subsequent modulations may still have degradation, resulting in inferior performance, as shown in Table. 4 [configuration d)] and Fig. 6.

Facial Component Loss. We compare the results of 1) removing all the facial component loss, 2) only keeping the component discriminators, 3) adding extra feature matching loss as in [62], and 4) adopting extra feature style loss based on Gram statistics [15]. It is shown in Fig 7 that component discriminators with feature style loss could better capture the eye distribution and restore the plausible details.

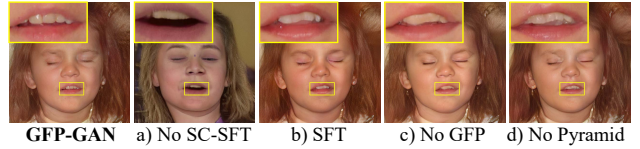


Figure 6: Ablation studies on CS-SFT layers, GFP prior and pyramid restoration loss. **Zoom in for best view.**

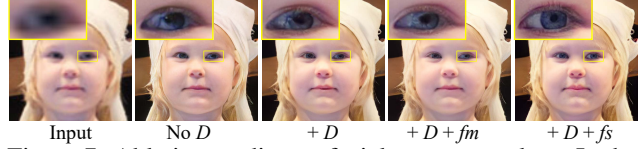


Figure 7: Ablation studies on facial component loss. In the figure, D , fm , fs denotes component discriminator, feature



Figure 8: Results on dark-skinned faces.



Figure 9: Limitations of our model. The results of PSFRGAN [6] are also presented.

4.4. Discussion and Limitations

Training bias. Our method performs well on most dark-skinned faces and various population groups (Fig. 8), as our method uses both the pretrained GAN and input image features for modulation. Beside, we employ reconstruction loss and identity preserving loss to restrict the outputs to retain fidelity with inputs. However, when input images are gray-scale, the face color may have a bias (last example in Fig. 8), as the inputs do not contain sufficient color information. Thus, a diverse *and* balanced dataset is in need.

Limitations. As shown in Fig. 9, when the degradation of real images is severe, the restored facial details by GFP-GAN are twisted with artifacts. Our method also produces unnatural results for very large poses. This is because the synthetic degradation and training data distribution are different from those in real-world. One possible way is to learn those distributions from real data instead of merely using synthetic data, which is left as future work.

5. Conclusion

We have proposed the GFP-GAN framework that leverages the rich and diverse generative facial prior for the challenging blind face restoration task. This prior is incorporated into the restoration process with channel-split spatial feature transform layers, allowing us to achieve a good balance of realness and fidelity. Extensive comparisons demonstrate the superior capability of GFP-GAN in joint face restoration and color enhancement for real-world images, outperforming prior art.

表4: 在CelebA-Test数据集上进行盲人脸恢复时的消融实验结果。

Configuration	LPIPS↓	FID↓	NIQE↓	Deg.↓
Our GFP-GAN with SC-SFT	0.3646	42.62	4.077	34.60
a) No spatial modulation	0.550 (↑)	60.44 (↑)	4.183 (↑)	74.76 (↑)
b) Use SFT	0.387 (↑)	47.65 (↑)	4.146(↑)	34.38 (↓)
c) w/o GFP	0.379 (↑)	48.47 (↑)	4.153 (↑)	35.04 (↑)
d) – Pyramid Restoration Loss	0.369 (↑)	45.17 (↑)	4.284 (↑)	35.50 (↑)

事实（尤其是在图5中的WebPhoto-Test部分）表明，GFP-GAN在头发和耳朵等面部特征上的表现也优于其他方法。这是因为GFP先验考虑的是整个脸部而非单独的部分。借助SC-SFT层，我们的模型能够实现高保真度。如图5的最后一行所示，大多数先前的方法无法恢复闭合的眼睛，而我们则能成功地以较少的伪影进行还原。

4.3. 消融实验

CS-SFT 层。如表4 [配置a)]和图6所示，当我们移除空间调制层*i.e.*，仅保留不包含空间信息的潜在代码映射时，即使使用了保持身份损失（高LIPS评分和大Deg.），恢复的人脸也无法保留面部识别特征。因此，CS-SFT 层中使用的多分辨率空间特征对于保持保真度至关重要。当我们把 CS-SFT 层切换为简单的 SFT 层 [表4的配置b)]时，我们观察到：1) 所有指标上的感知质量下降；2) 保留了更强的身份信息（Deg.更小），因为输入图像特征对所有调制后的特征施加影响，并且输出偏向于退化的输入，从而导致更低的感知质量。相比之下，CS-SFT 层通过调节部分特征提供真实感和保真度的良好平衡。

预训练GAN作为GFP。预训练GAN提供了丰富多样的特征用于恢复。如果不使用生成面部先验，则如表4 [配置c)]和图6所示，性能会下降。

金字塔恢复损失。金字塔恢复损失被应用于降质去除模块中，增强了对现实世界复杂降质的恢复能力。如果没有这一中间监督机制，在后续调制中的多分辨率空间特征可能仍然存在降质问题，从而导致性能不佳，如表4 [配置d)] 和图6所示。

面部组件损失。我们比较了以下四种情况的结果：1) 移除所有面部组件损失，2) 仅保留组件判别器，3) 如文献[62]所述添加额外的特征匹配损失，4) 基于Gram统计量采用额外的特征风格损失[15]。图7显示，带有特征风格损失的组件判别器能够更好地捕捉眼睛分布并恢复合理的细节。

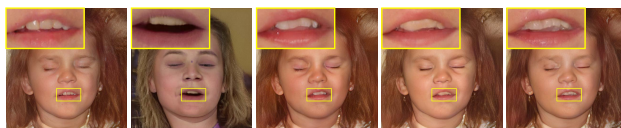


图6: CS-SFT层、GFP先验及金字塔恢复损失的消融研究。放大查看最佳效果。

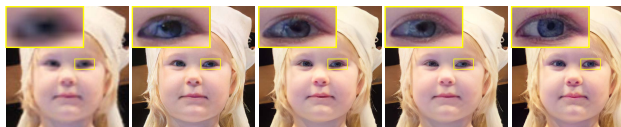


图7: 面部组件损失的消融研究。在图中， D 、 fm 、 fs 表示组件判别器和特征



Figure 8: Results on dark-skinned faces.

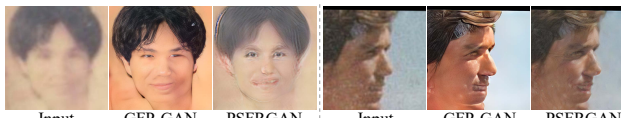


图9: 我们的模型的局限性。同时也展示了PSFR-GAN[6]的结果。

4.4. 讨论与限制

训练偏差。我们的方法在大多数肤色较深的脸部图像和各种人群中表现良好（见图8），因为我们的方法同时使用了预训练的生成对抗网络（GAN）和输入图像特征进行调制。此外，我们采用重构损失和身份保持损失来限制输出以保留与输入的一致性。然而，当输入图像为灰度图像时，面部颜色可能会出现偏差（见图8中的最后一个例子），因为输入中不包含足够的色彩信息。因此，需要一个多样且平衡的数据集

限制。如图9所示，当真实图像的退化严重时，GFP-GAN恢复的人脸细节会出现扭曲和伪影。我们的方法对于非常大的姿态变化也会产生不自然的结果。这是因为合成退化和训练数据分布与现实世界中的情况不同。一种可能的方法是从真实数据中学习这些分布，而不仅仅是使用合成数据，这留待未来的工作来完成。

5. 结论

我们提出了一种基于GFP-GAN框架的方法，该方法利用了丰富且多样化的生成面部先验知识，以应对具有挑战性的无参考人脸修复任务。通过在修复过程中引入通道分割的空间特征变换层，我们将这种先验知识融入到了恢复过程之中，从而实现了真实感和保真度的良好平衡。大量的对比实验表明，GFP-GAN在实际图像的人脸修复与色彩增强方面表现出卓越的能力，并且优于现有的方法。

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